Dr. JAEHYUK LEE, Ph.D.

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RESEARCH INTEREST

I am interested in computer systems security, especially in: hardware/software codesign for security, Trusted Execution Environment (TEE), secure IO design for TEE, software/hardware vulnerability research, side-channel attacks and mitigations, and fuzzing

TECHNICAL SKILLS

Software: C, C++, Rust, Scala, Python, TF-A, TF-RMM, Caliptra FW, PSC FW (NVIDIA IROT), Linux Kernel, KVM, QEMU, ARM/X86 Assembly, Fuzzing, GDB, CoCLI

Hardware: GEM5 (x86 simulator), Processor Pipeline, Cache, TLB, Microcode, HW Side Channel, (System)Verilog, Chisel (RISC-V), Intel SGX & TDX, ARM CCA & TrustZone, IOMMU, PCIe

Spec: Entity Attestation Token (EAT), Concise Reference Integrity Manifest (CoRIM), DICE Protection Environment (DPE), Security Protocol and Data Model (SPDM), TEE Device Interface Security Protocol (TDISP)

WORK EXPERIENCE

Senior Security Engineer: NVIDIA Corporation, Santa Clara, CA **Jul 2024 – Current**

- Owned device evidence and CoRIM generation for Vera and Grace confidential computing
  - Implemented cross-layer firmware support across *RMM, ATF, PSCFW (IROT), and Caliptra DPE* to enable end-to-end *Vera CCA* evidence generation.
  - Implemented the software stack to enable *Baremetal Secure AI* attestation evidence generation for Vera and Grace CPUs.
  - Developed CoRIM for Vera CCA and Vera/Grace baremetal attestation
- Owned Live Firmware Activation (LFA) support for RMM
  - Built the supporting software stack across *RMM, ATF, and Vera IROT* to enable live firmware activation.
- Experienced in the end-to-end SBIOS firmware development lifecycle, including build, validation, testing, and firmware package releases.
- Engineered secure tenant data loading (e.g., ML models) for NVFlare by integrating encrypted disk and early attestation key provisioning directly into initramfs, securing the boot sequence

Postdoc Researcher: Georgia Institute of Technology, Atlanta, GA **Jan 2022 – Jul 2024**

- Developed an automated fuzzer selection tool that dynamically selects fuzzer(s) based on runtime analysis [4]. Anyone having no knowledge of fuzzers can simply utilize it for automate vulnerability discovery
- Experienced in vulnerability research on Intel and ARM's latest hardware feature designed for confidential VM (TDX & CCA). Identified new side-channel issues related to resource allocation in the construction of confidential VMs
- Implemented a secure IO path between confidential VMs and peripherals such as GPUs by leveraging isolation guarantees of IOMMU (SMMU) and ARM CCA, avoiding encryption overhead [1]

Postdoc Researcher: Korea Advanced Institute of Science and Technology, Korea **Mar 2021 – Dec 2021**

- Implemented cache side-channel attack on X86 and Apple M1 chips, unveiling the memory access pattern without evicting the victim's cache entries and *completely bypassing side-channel defenses* monitoring cache evictions [10]
- Took the lead in managing and mentoring a group of 20 Ph.D. students

Research Intern: Georgia Institute of Technology, Atlanta, GA **Nov 2019 – May 2020**

- Designed and implemented a *hardware-based side-channel mitigation* on the GEM5 simulator, enabling user-space applications to subscribe to micro-architectural events (cache and TLB evictions) [3]

Research Intern: Microsoft Research, Redmond, WA **May 2016 – August 2016**

- Researched memory corruption vulnerabilities in Intel SGX and executed a practical ROP attack against encrypted binaries, demonstrating compromise of confidentiality guarantees [5].
- Developed a secure CDN that enables enterprises to offload sensitive user data to untrusted third-party cloud providers without compromising confidentiality.

PUBLICATION & PATENT (826 citations)

top-tier conferences (SP, USENIX Security, NDSS) and Journal (TDSC)

[1] **PORTAL: Fast and Secure Device Access with Arm CCA for Modern Arm Mobile SoC** | Paper **SP 2025**

- TEEs designed for confidential VMs face limitations in directly interfacing with peripherals, necessitating encryption and decryption overhead when transmitting data to devices. Portal addresses this concern by implementing access control
- Redesigned IOMMU subsystem within the Linux kernel and migrated the resource allocation component, including SMMU page table management, to TF-RMM. Introduced RMI interface calls between KVM and TF-RMM to enable confidential VM to have a dedicated, isolated DMA region.

[2] **Survey On DeFi Users' Perception of Security Breaches and Countermeasures** | Paper **USENIX 2024**

- Investigated DeFi users' security perceptions and commonly adopted practices, and how those affected by previous scams or hacks (DeFi victims) respond and try to recover their losses.
- Conducted qualitative analysis through coding 14 interviewers' responses using Dedoose.

[3] **SENSE: Enhancing Microarchitectural Awareness via Subscription-Based Notification** | Paper **NDSS 2024**
- SENSE grants TEEs access to micro-architectural information, facilitating preemptive protection against side-channel attacks targeting caches and TLBs. SENSE is implemented using the GEM5 system-level processor simulator.
- Modified GEM5 out-of-order processor pipeline to redirect its execution to the pre-registered handler by implementing micro-code. I modified cache and TLB design to introduce additional bits tracking events with minimal cost.

[4] **autofz: Automated Fuzzer Composition at Runtime** | Paper | Hacker News **USENIX 2023**
- Autofz alleviates the need for users to manually configure the best fuzzer for each specific target. Similar to AutoML, it automatically and adaptively configures best-performing fuzzers at runtime and achieves the best performance.
- Experienced in multiple fuzzers such as LearnAFL, RedQueen, MOpt, AFLFast, AFL, LAF-Intel, Angora, FairFuzz, Radamsa, QSYM.

[5] **Hacking in Darkness: Return-oriented Programming against Secure Enclaves** | Paper | Talk | Demo **USENIX 2017**
- Dark-ROP demonstrates practical exploitation of memory corruption vulnerabilities within enclaves through Return-Oriented Programming (ROP), utilizing special oracles induced by the design of Intel SGX.
- Located ROP gadgets by leveraging the side effects of newly introduced instructions for Intel SGX, all without prior knowledge of the binary details of the target process. Implemented full ROP chain and demonstrated attack.

[6] **SGX-Bomb: Locking Down the Processor via Rowhammer Attack** | Paper | Hacker News | Demo **SysTEX 2017**
- SGX-Bomb showcases how Rowhammer attacks can jeopardize enclave integrity, posing a risk of denial-of-service (DoS) scenarios, which is especially concerning for cloud providers that host untrusted enclave programs.
- Implemented kernel driver to reverse engineer mapping between physical address to memory bank for locating two adjacent rows. Demonstrated rowhammer attack inside Intel SGX with X86 assembly and C++ coding.

[7] **PrivateZone: Providing a Private Execution Environment using ARM TrustZone** | Paper **TDSC 2016**
- PrivateZone provides a secure TEE for developers to deploy and execute user processes on mobile platforms. It delivers assurances similar to TrustZone without imposing additional costs on developers to deploy applications
- Implemented stage-2 page table for Privatezone and developed the logic for context switching, effectively managing the transition between trusted stage-2 page tables and untrusted ones. Experienced in ARM assembly coding.

[8] **OpenSGX: An Open Platform for SGX Research** | Paper | Wikipedia: SGX **NDSS 2015**
- OpenSGX enables Intel SGX exploration with instruction-level emulation, addressing limitations in TEE software development and offering a comprehensive ecosystem, including an OS, enclave handling, user libraries, and debugging
- Implemented X86 assembly facilitating context switching within the SGX environment, enabling the transition between entering and exiting the SGX. Developed stub-call allowing SGX to invoke system calls through a customized libc

[9] **A Rule Extraction Method Using Relevance Factor for FMM Neural Networks** | Paper **KTSDE 2013**
- Enhanced Fuzzy Min-Max (FMM) neural network by incorporating the frequency of features in training.
- Resolved ambiguity by prioritizing clarity and relevance, while ensuring distinctiveness between features and patterns.

[10] **PRIME+RETOUCH: When Cache is Locked and Leaked** | Paper **Arxiv**
- PRIME+RETOUCH attack circumvents cache-based side-channel defense monitoring eviction events by utilizing cache replacement policy metadata. The attack is demonstrated on both Intel x86 and Apple M1 architectures.
- Conducted reverse engineering to analyze the cache eviction policy of the most recent X86 processor and the Apple M1 chip. Developed PoC in X86 and ARM assembly, employing pointer chasing along with memory barriers

[11] **Apparatus and method for open and private IoT gateway using Intel SGX** | Patent **KR102162018B1**
- The invention introduces a secure IoT gateway design, utilizing Intel SGX for protocol conversion and encrypted data transfer. It includes enclave designs for cross-platform communication, encrypted key management, and secure updates.

SOFTWARE ARTIFACTS

Autofz:	https://github.com/sslab-gatech/autofz	13	★85
OpenSGX:	https://github.com/sslab-gatech/opensgx	82	★308
SGXBomb:	https://github.com/sslab-gatech/sgx-bomb	4	★18

EDUCATION

Institution	**Degree**	**Field of Study**	**Dates**	**GPA**
Korea Advanced Institute of Science and Technology	*Ph.D.*	Information Security	Aug 2015 - Feb 2021	3.95
Korea Advanced Institute of Science and Technology	*M.S.*	Information Security	Sep 2013 - Aug 2015	4.1
Handong Global University	*B.S.*	Computer Science	Mar 2010 - Aug 2013	4.2

PROFESSIONAL ACTIVITIES

[1] **Journal Review**
- IEEE Transactions on Dependable and Secure Computing — Impact Factor: 7.3, *2024*
- Computer & Security — Impact Factor: 5.7, *2024*
- Journal of Information Security and Applications — Impact Factor: 5.6, *2024*
- IEEE Access — Impact Factor: 3.9, *2024*
- ACM Transactions on Internet Technology — Impact Factor: 3.9, *2024*

- IEEE Journal of Communications and Networks — Impact Factor: 3.6, *2024*

[2] **Conference Review**

- IEEE Mobile Ad Hoc and Smart Systems (MASS) — *2024*
- IEEE International Conference on Parallel and Distributed Systems (ICPADS) — *2024*

[3] **External Reviewer**

- IEEE Symposium on Security and Privacy (Oakland) — *2016, 2017, 2018, 2020, 2021, 2022*
- USENIX Security Symposium (Security) — *2015, 2021, 2022, 2023, 2024*
- ACM Conference on Computer and Communications Security (CCS) — *2015, 2017, 2018, 2019, 2023*
- Network and Distributed System Security Symposium (NDSS) — *2019, 2020, 2021*
- ACM ASIA Conference on Computer and Communications Security (ASIACCS) — *2015, 2016, 2018*
- ACM Symposium on Operating Systems Principles (SOSP) — *2021*
- ACM International World Wide Web Conference (WWW) — *2018*

INVITED TALK

[1] **A Novel Cache Side Channel Attack without Attacker Eviction**

- Presented at Intel Side Channel Academic Program (SCAP), Virtual, Sep 2020

[2] **Hacking in Darkness: Return-oriented Programming against Secure Enclaves**

- Presented at the 26th Usenix Security | Presentation Link, Vancouver, BC, Canada, Aug 2017
- Presented at KimchiCon | Presentation Link, Daejeon, South Korea, Aug 2017

[3] **Tutorial: Trusted Execution Environment: TrustZone, ITP and SGX**

- Korea Institute of Information Security and Cryptology, Seoul, South Korea, Apr 2015

THESIS

Unintended Consequences of System Design: Unpremeditated Usage of Benign System Components Devastates Security — Ph.D. Thesis.

Detecting Emulated Environment: Exploiting Behavioral Discrepancies In QEMU — M.S. Thesis.